

Diseases of Bones, Joints and Muscles

Learning Objectives

- ⇒ To know about brief introduction about Musculo skeletal diseases
- ⇒ To know about the clinical features of osteoarthritis, osteoporosis, frozen shoulder syndrome, calcaneal spur, plantar fasciitis, tennis elbow, carpal tunnel syndrome, muscular diseases, muscular dystrophy
- ⇒ To know about necessary physical examinations to be performed in above mentioned diseases
- ⇒ To know about the laboratory investigations preferred in above mentioned diseases and interpretations of these investigations

Osteoarthritis

Osteoarthritis (see Figure 189) (OA) is a chronic degenerative inflammation of joints characterized by disruption and partial loss of joint cartilage along with other joint changes, including bone hypertrophy (osteophyte formation). Symptoms include gradually developing pain aggravated or triggered by activity, stiffness lasting < 30 minutes on awakening and after inactivity and occasional joint swelling. Diagnosis is confirmed by x rays of the affected joint.

- ⊛ This is the most common joint disorder, often becomes symptomatic in the 40s and 50s of the age group and after the age of 80, this is universal in presentation. Only ½ of the patients with pathological changes of OA have symptoms. Below 40 years, most OA

is seen in males and results from trauma. Women predominate from age between 40 and 70.

- ⊛ OA is classified as primary (idiopathic) and secondary (due to other clinical conditions such as trauma or congenital joint diseases or infection of the joints or endocrine or neurological diseases)
- ⊛ Onset is gradual, usually beginning with one or few joints. Pain is the earliest symptom, sometimes worsened by weight bearing and relieved by rest, but can eventually become constant. Stiffness follows awakening or inactivity and pain may persist for < 30 minutes and lessens with movement. As OA progresses, the degree of movement of afflicted joint will be reduced and tenderness and crepitus or grating sensations develop.

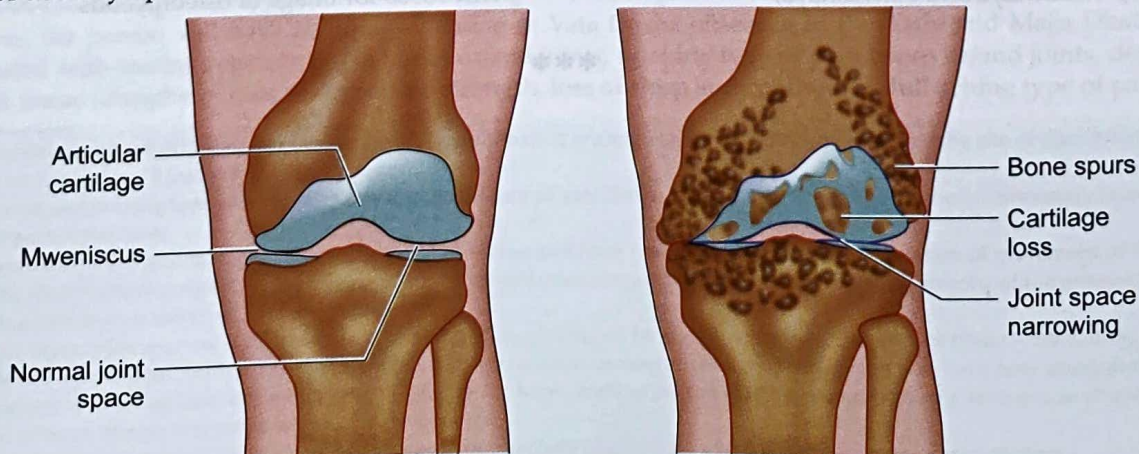


Figure 189: Osteoarthritis



Figure 190: X ray showing normal and abnormal knee joint

- ❖ Proliferation of the cartilage, bone, ligament, tendon, capsule and synovium along with varying amount of joint effusion, ultimately causing joint enlargement. Flexion contractures may eventually develop.
- ❖ Tenderness on palpation and pain on passive motion are relatively late signs of OA. Deformities and subluxations may develop during the progression of the disease.
- ❖ Diagnosis is made by performing X-ray (see **Figure 190**) of the affected joint.

Osteoporosis

Osteoporosis (see **Figure 191**) is a progressive, metabolic bone disease that decreases bone density (bone mass per unit volume), with deterioration of bone structure.

Skeletal weakness leads to fractures with minor or inapparent trauma, particularly in the thoracic and lumbar spine, wrist and hip.

Acute or chronic back pain is common. Diagnosis is by dual energy x ray absorptiometry.

- ❖ Normally bone formation and resorption are closely coupled. Osteoblasts (cells that make the organic matrix of the bone and then mineralize bone) and osteoclasts (cells that resorb bone) are regulated by parathyroid hormone, calcitonin, oestrogen, vitamin D and other local factors such as prostaglandins.
- ❖ Osteoporotic bone loss affects cortical and trabecular (cancellous) bone. Cortical thickness and the number and size of trabeculae decrease, resulting in increasing porosity. Trabeculae may be disrupted or entirely absent.

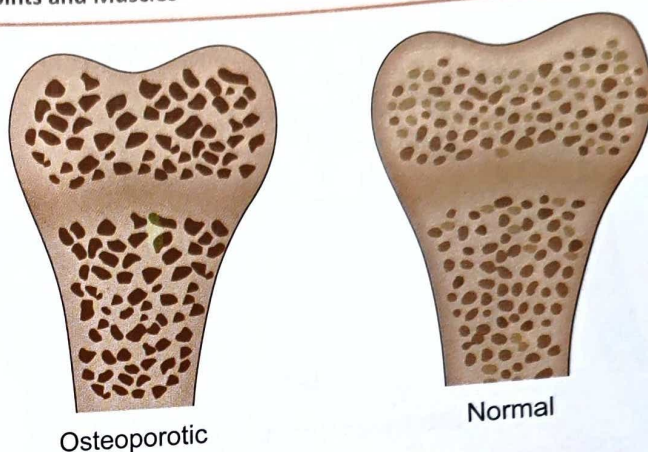


Figure 191: Osteoporosis

- ❖ Osteoporosis can develop as primary (idiopathic) or secondary (due to some other factors) based on the nature of exposure to the aetiological factor.
- ❖ In primary type, cause is unidentifiable, and in secondary type, the disease may develop due to cancer or COPD or chronic renal failure or endocrine disease such as Cushing's syndrome/ hyperparathyroidism/ diabetes mellitus/ hyperthyroidism or hypervitaminosis A or liver disease or rheumatoid arthritis (RA) or sarcoidosis.
- ❖ Most of the chronic pain typical of osteoporosis results from fractures, which may develop after minimal, inapparent or no trauma.
- ❖ Patient may be asymptomatic for years until fractures begin to occur.
- ❖ Eventually often patients develop pain in the bones or muscles, particularly of the back. Vertebral compression fractures are common, usually in weight bearing vertebrae (T6 and below).
- ❖ Onset of the pain is sudden, usually does not radiate, aggravated by weight bearing, may cause local tenderness, and generally begin to subside in 1 week. However, residual pain may last for months or be constant.
- ❖ The person may have dorsal kyphosis due to multiple thoracic compression fractures, with exaggerated cervical lordosis (dowager's lump). Abnormal stress on spinal muscles and ligaments may cause chronic, dull, aching pain, particularly in the lower back. Fractures can develop at other sites, commonly the hip or wrist, usually from falls.
- ❖ Diagnosis is done by Dual Energy X-Ray Absorptiometry (DEXA).

Frozen Shoulder Syndrome

Frozen shoulder (see **Figure 192**) is a clinical condition where in the movements of the affected shoulder (glenohumoral joint) become limited associated with severe pain and increased stiffness.

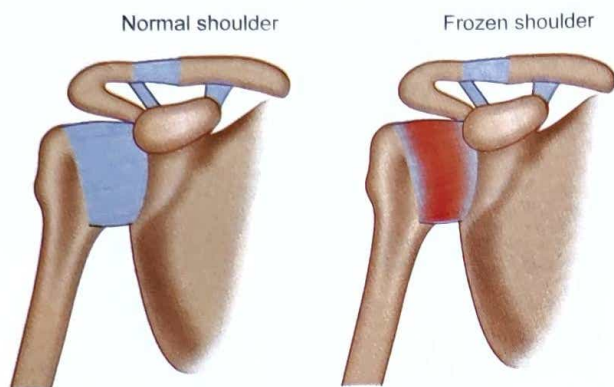


Figure 192: Frozen shoulder syndrome

- ❖ The patient most often gives the history of trauma.
- ❖ The underlying pathology is that an exudate causes the layers of the infra-articular synovial membrane (inferior aspect of the synovial membrane) and the capsule to adhere to one another. Patients between 45 – 60 years are commonly affected. This is more commonly seen in females.
- ❖ The patient experiences pain which becomes worse at night and prevents the patient from sleeping on the affected side. Gradual stiffness of the shoulder follows. In a matter of months, all movements especially abduction and external rotation are restricted. As the process continues, the pain abates. By this time, the muscles around the shoulder show signs of disuse atrophy. Months later, stiffness gradually lessens and the disease recovers spontaneously.
- ❖ 3 phases of the disease can be observed: Each phase lasts 4 to 8 months and the whole process takes about 2 years.
 - Increasing pain and increasing stiffness
 - Decreased pain with persistent stiffness
 - Disappearance of stiffness with return of all the movements
- ❖ Diagnosis is done by physical examination and X-ray

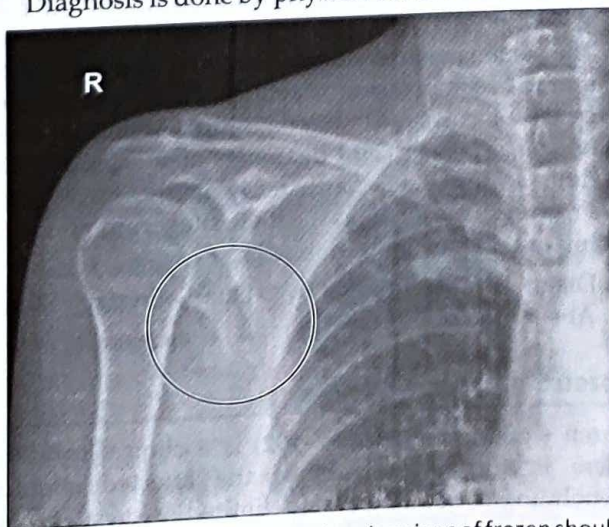


Figure 193: X ray right shoulder showing signs of frozen shoulder

Calcaneal Spur

Calcaneal spur is a bony projection forwards from the undersurface of the calcaneal tuberosity. It is nothing but ossification of the plantar fascia at its calcaneal end. This has very little significance so far as the pain in the heel is concerned. This means if a patient complains of pain in the heel and on x ray, one can find the presence of calcaneal spur, the clinician cannot infer that the calcaneal spur is the cause of pain. Very often, inflammation of the soft tissue or bursa beneath the spur gives rise to pain.

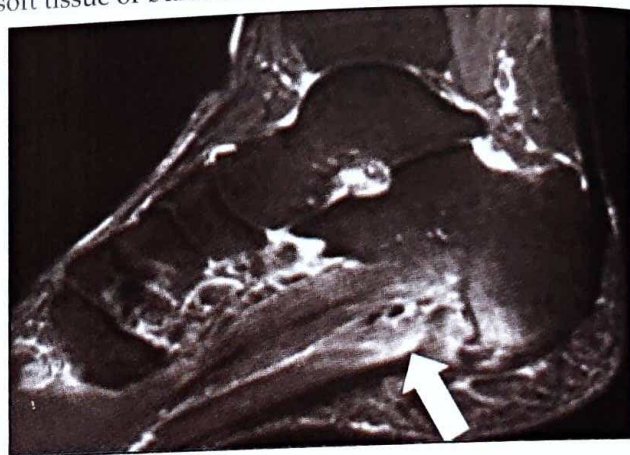


Figure 194: X ray foot showing calcaneal spur

- ❖ Calcaneal spur is often associated with plantar fasciitis (painful inflammation of the fibrous band of connective tissue, i.e. plantar fascia). These may develop on the inferior surface of the heel, that occurs over a period of many months. These are caused by strains on the foot muscles and ligaments, stretching of the plantar fascia, and repeated tearing of the membrane that covers the calcaneus. These are common among athletes whose activities include large amounts of running and jumping.
- ❖ This disease will not cause any major symptoms. This may be associated with intermittent or chronic pain, especially while walking, jogging or running. The character of pain will be like knife or pin sticking into the bottom of the feet, if the inflammation develops at the point of spur formation.
- ❖ Diagnosis is done by x ray. As far as prognosis is concerned, 20 to 75% of the patients no longer have any symptoms within a maximum of one year after the onset of the symptoms.

Plantar Fasciitis

Plantar fasciitis, a painful inflammation of the fibrous band of connective tissue (plantar fascia) that runs along the bottom of the foot and connects the heel bone to the ball of the foot. In this condition, the person will have unbearable pain under heel particularly during walking when the weight of the body is carried by the heel. The cause seems to be a small tear in the attachment of plantar

fascia to the os calcis. Non specific infection from non specific urethritis or from specific gonococcal infection may develop this condition. Sometimes, a bony spur may be seen at the attachment of the plantar fascia, i.e. calcaneal spur. This may or may not be the cause of pain.

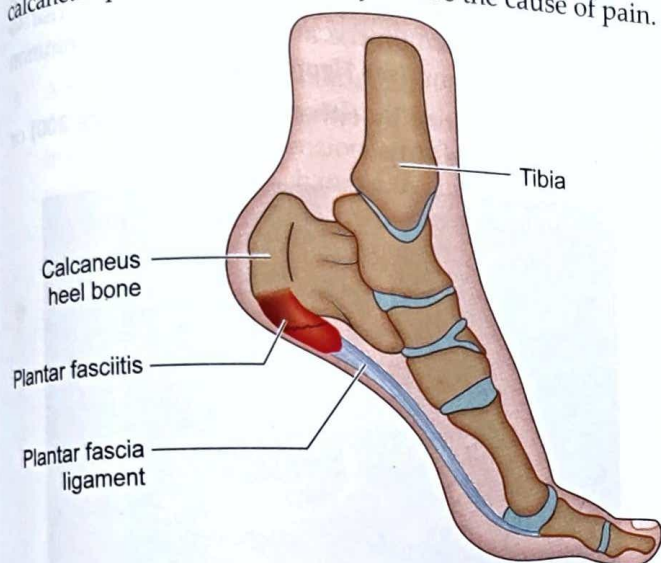


Figure 195: Plantar fasciitis

- Plantar fasciitis is an inflammation of the fibrous tissue (plantar fascia) along the bottom of foot that connects heel to the toes.
- The cause of plantar fasciitis is unknown. This is more commonly seen in runners or the people having overweight or obesity.
- **Symptoms:** This is associated with intense throbbing type of pain often occurs with first steps in the morning which gets relieved after waking up and walking for few steps. This may recur standing for a long period of time. Most of the occasions, this is unilateral, but bilateral presentation may also be seen.

- On examination, tenderness of the foot at the heel on palpation is observed which may worsen on dorsiflexion of the foot.
- Diagnosis is done by either X-ray (see Figure 194) or MRI (see Figure 196).



Figure 196: MRI showing plantar fasciitis

Tennis Elbow

Tennis elbow is a clinical condition where the patient complains of pain in the lateral epicondyle of the humerus where the extensor group of muscles of forearm is attached. This is also called as lateral epicondylitis. This is more often seen among the tennis players and hence the name, but quite a number of sufferers of this condition have never played tennis. The pain is aggravated when the patient tends to extend the wrist and the fingers, more so against resistance. Diagnosis is done by x ray or MRI.

- The major cause of the disease is repetitive, excess, strained use of the extensor group of the muscles of the forearm which is attached to the lateral epicondyle of the forearm.

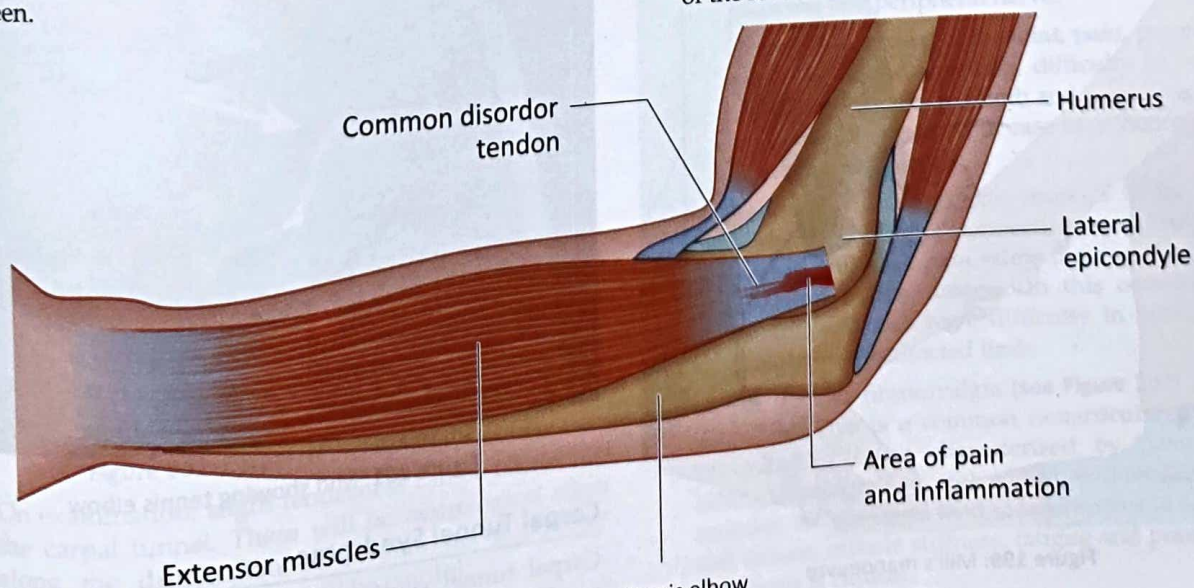
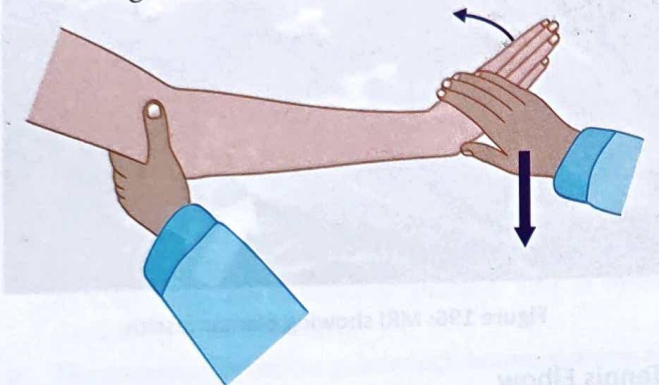


Figure 197: Tennis elbow

- ❖ Symptoms include pain, burning or ache along the lateral side of the forearm and elbow. It is worsened by continuous extension activity of the forearm and which spread even to the wrist. The grip may become weak.

- ❖ On examination, localized tenderness is felt on the lateral epicondyle where the extensor group of muscles of the forearm is attached. Cozen's test is performed and Mill's manoeuvre is done to confirm the diagnosis of the disease.



Cozen's Test

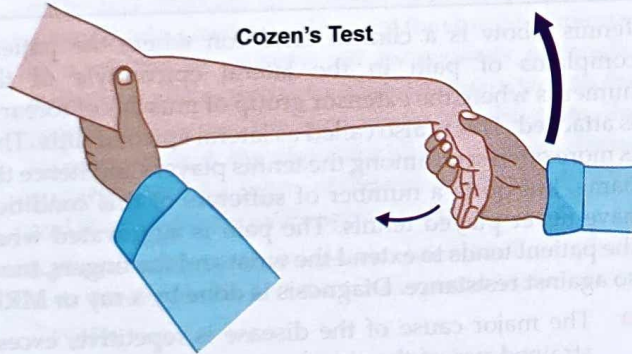


Figure 198: Cozen's test

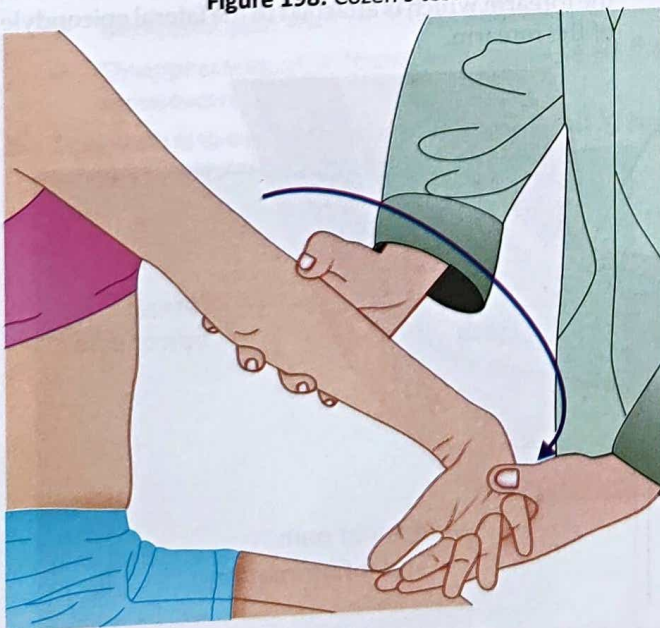


Figure 199: Mill's manoeuvre

- **Cozen's test:** The patient is asked to extend his clenched fist against resistance, considerable pain is experienced at the lateral epicondyle. (see Figure 198)
- **Mill's manoeuvre:** Patient's wrist is passively flexed when his forearm is pronated. This gives rise to tremendous pain on the attachment of the common extensor tendons. (see Figure 199)
- ❖ Diagnosis is done by either X ray (see Figure 200) or MRI (see Figure 201).



Figure 200: X ray showing tennis elbow

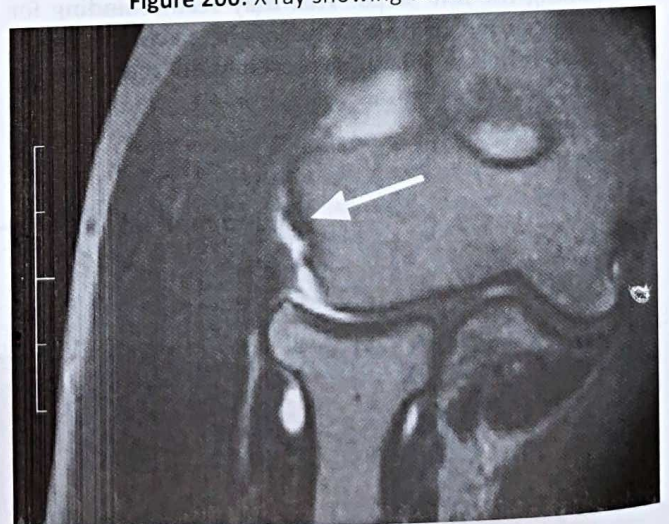


Figure 201: MRI showing tennis elbow

Carpal Tunnel Syndrome

Carpal tunnel syndrome (see Figure 202) is a clinical condition which is responsible for the compression of the carpal tunnel. This condition is an example of stenosing

tenovaginitis. Here, the cause is not only thickening of the flexor retinaculum, but involvement of the synovial sheaths of the flexor tendons or dislocation of lunate bone canal, median nerve in particular. The main presenting complaint is difficulty in flexing the fingers.

- A wrist fracture or inflammation of the wrist joint in the patients of rheumatoid arthritis or activities that involve extreme extension and flexion of hand and wrist or hormonal changes during pregnancy may cause the compression of the carpal tunnel, which compresses median nerve.
- The main complaint is some sort of difficulty in flexing the fingers with pain and neurological deficits of the median nerve, e.g. paraesthesia or numbness or incoordination or weakness of the muscles innervated by median nerve. The pain disturbs the sleep of the patient. Occasionally shock like sensations that radiate to the thumb and index, middle and ring fingers, pain may radiate upto forearm towards the shoulder. Weakness and clumsiness of the hand which makes the person to perform finer activities with difficulty such as buttoning the clothes.

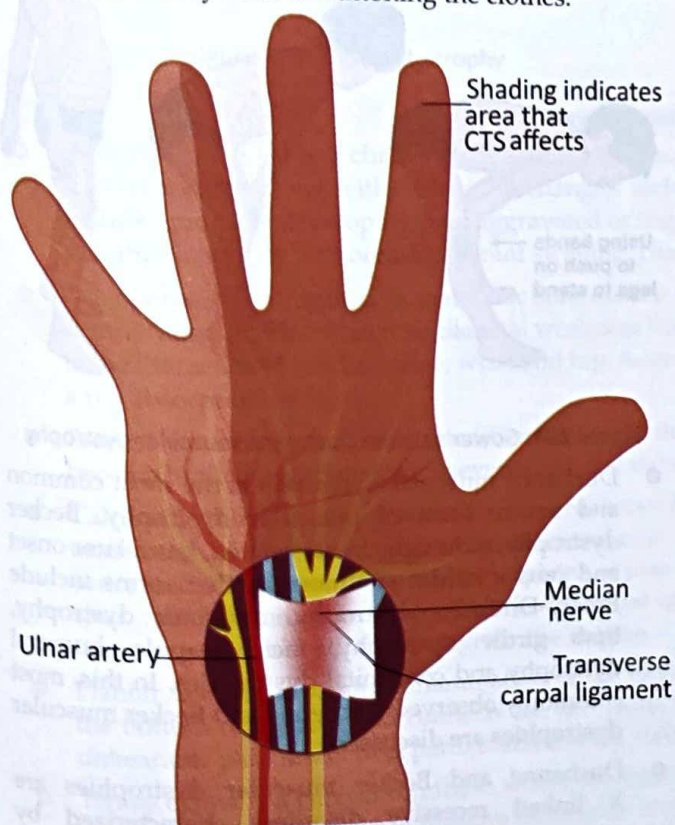


Figure 202: Carpal tunnel syndrome

- On examination, slight tenderness can be elicited on the carpal tunnel. There will be neurological signs along the distribution of median nerve. Flexion movement of the fingers will be painful.
- Diagnosis is done by nerve conduction study of median nerve which demonstrate a delay in the carpal tunnel.

Muscular Diseases

This group of diseases must be considered while making the differential diagnosis related to bones and joints. Trauma, fracture and fibromyalgia are most common clinical presentation of the musculoskeletal disorders. The patient may have presentation of severe pain which results in the restriction of movement of the affected part of the body.

- The role of physical examination include to ascertain the structures involved, the nature of underlying pathology, functional consequences of the process and the presence of systemic and extraarticular manifestations. Whenever, analysing muscular pain, one should rule out the affliction of bone, joint, muscle, bursa, ligament and tendon.
- Bony pain is often described as penetrating, boring or deep pain, characteristically worse at night. Localized bony pain may occur in tumour, chronic infection (osteomyelitis), avascular necrosis or osteoid osteoma (a benign bony tumour).
- Pain arising from the joint can be well localized, worsens on performing the movement of the affected joint.
- Muscular pain is usually described as stiffness and is poorly localized, and aggravated by use of the muscle. It is sometimes associated with muscular weakness such as poliomyositis. Partial muscular tears may be painful, but complete rupture is relatively pain free.
- Fracture pain is sharp and stabbing, aggravated by attempting the movement or use and relieved by rest and splintage.
- Shooting pain often caused by mechanical trapping of a peripheral nerve.
- Bruising, looseness of the joint, pain, popping or snapping sound, swelling, difficulty in bearing the weight on affected limb and weakness of the joint may be observed in case of affliction of the ligament of the body.
- Tendons will connect the muscles to the bones which are fibrous connective structures. The person will have pain while moving the muscle involved in the process. In this occasion, the person will also have difficulty in bearing the weight on the affected limb.
- In this section, fibromyalgia (see Figure 203) will be discussed. This is a common nonarticular disorder of unknown cause characterized by generalized aching (sometimes severe); widespread tenderness of muscles, areas around tendon insertions and adjacent soft tissues; muscle stiffness; fatigue and poor sleep. Diagnosis is clinical.

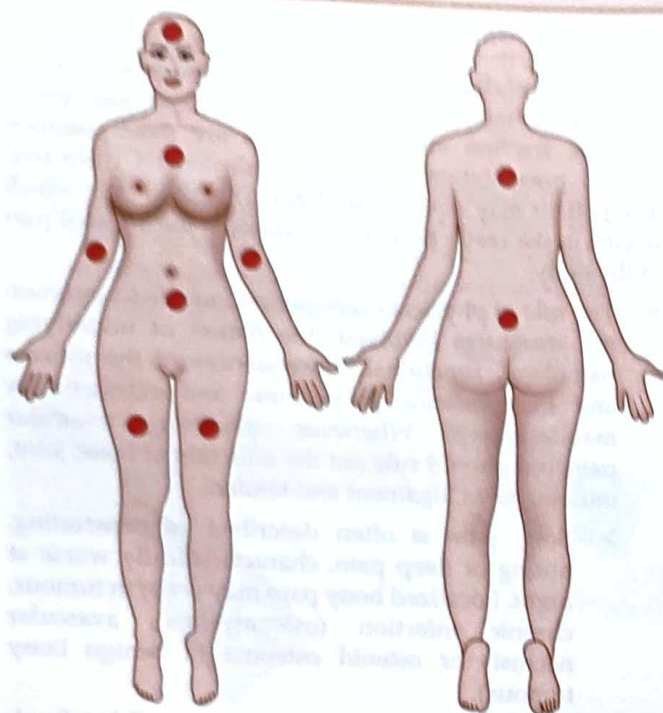


Figure 203: Fibromyalgia

- In this disease, any fibromuscular tissues may be involved, especially those of the occiput, neck, shoulders, thorax, low back and thighs. There is no specific histologic abnormality. It is 7 times more common in females, usually young or middle aged women, but can occur in men, children and adolescents. Because of gender difference, it is often overlooked in males. Sometimes, it occurs in patients with systemic rheumatic symptoms.
- The cause is unknown, but disruption of stage 4 sleep may contribute, as can emotional stress. Patients may tend to be perfectionists.
- Symptoms and signs are generalized, in contrast to localized soft tissue pain and tenderness. Stiffness and pain frequently begin gradually and diffusely and have an achy quality for a period of at least 3 months. Symptoms can be exacerbated by environmental or emotional stress, poor sleep, trauma or exposure to dampness or cold.
- Patients tend to be stressed, tense, anxious, fatigued, ambitious and sometimes depressed. Many patients have irritable bowel syndrome symptoms or migraine or tension headaches. Pain may worsen with fatigue, muscle strain or overuse. Specific discrete areas of muscle (tender points) may be tender when palpated. To do the examination, the clinician has to do digital palpation with a force of about 4 kg.
- Diagnosis is clinical.
 - Generalized pain and tenderness, especially if disproportionate to physical findings

- Negative laboratory results despite widespread symptoms
 - Fatigue as the predominant symptom
- As far as prognosis of the disease is concerned, this disease tends to be chronic but may remit spontaneously if stress reduces. It can also recur at regular intervals.

Muscular Dystrophy

Muscular dystrophies are inherited, progressive muscle disorders resulting from defects in one or more genes needed for normal muscular function. They are distinguished by the selective distribution of weakness and the specific nature of the genetic abnormality involved.

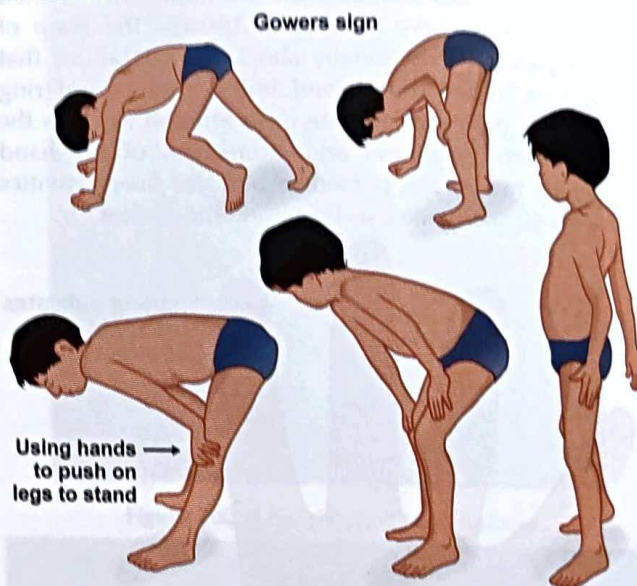


Figure 204: Gower's sign in Duchenne muscular dystrophy

- ① Duchenne muscular dystrophy is the most common and severe form of muscular dystrophy. Becker dystrophy, although closely related, has a later onset and causes milder symptoms. Other forms include Emery-Dreifuss dystrophy, myotonic dystrophy, limb girdle dystrophy, facio scapulo humeral dystrophy and congenital dystrophies. In this, most commonly observed Duchenne and Becker muscular dystrophies are discussed.
- ② Duchenne and Becker muscular dystrophies are X linked recessive disorders characterized by progressive muscular weakness caused by muscle fibre degeneration. Becker muscular dystrophy has later onset and causes milder symptoms. Diagnosis is suggested clinically and is confirmed by analysis of the protein product (dystrophin) of the mutated gene.
- ③ These dystrophies are caused by mutations at Xp21 focus.

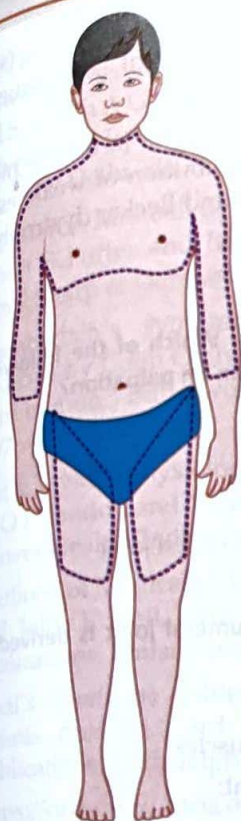


Figure 205: Becker dystrophy

- ⊛ Duchenne muscular dystrophy: manifests between ages 2 and 3 years. Weakness affects proximal muscles, typically in the lower limbs initially. Children frequently toe walk and have a waddling gait and lordosis.
- ⊛ They fall frequently and have difficulty running, jumping, climbing stairs and raising from the floor (Gower's sign).
- ⊛ Progressive weakness is steady, and limb flexion contractures and scoliosis develop.
 - Firm pseudohypertrophy (fatty and fibrous replacement of certain enlarged muscle groups, notably in the calves) develops.
 - Most children are confined to a wheelchair by 12 and die of respiratory complications by age of 20 years.
- ⊛ **Becker dystrophy (see Figure 205):** This disorder becomes typically symptomatic much later and is milder. Ambulation is usually preserved until at least age 15 and many children remain ambulatory into adulthood. Most affected children survive into their ages 30s and 40s.
- ⊛ Diagnosis is done by immunostaining analysis of dystrophin and DNA mutation analysis.

Key summary of the chapter

- ⊛ Osteoarthritis (OA) is a chronic degenerative inflammation of joints characterized by disruption and partial loss of joint cartilage along with other joint changes, including bone hypertrophy (osteophyte formation). Symptoms include gradually developing pain aggravated or triggered by activity, stiffness lasting < 30 minutes on awakening and after inactivity and occasional joint swelling. Diagnosis is confirmed by x rays of the affected joint.
- ⊛ Osteoporosis is a progressive, metabolic bone disease that decreases bone density (bone mass per unit volume), with deterioration of bone structure. Skeletal weakness leads to fractures with minor or inapparent trauma, particularly in the thoracic and lumbar spine, wrist and hip. Acute or chronic back pain is common. Diagnosis is by dual energy x ray absorptiometry.
- ⊛ Frozen shoulder is a clinical condition where in the movements of the affected shoulder (glenohumeral joint) become limited associated with severe pain and increased stiffness.
- ⊛ Calcaneal spur is a bony projection forwards from the undersurface of the calcaneal tuberosity. It is nothing but ossification of the plantar fascia at its calcaneal end. This has very little significance so far as the pain in the heel is concerned. This means if a patient complains of pain in the heel and on x ray, one can find the presence of calcaneal spur, the clinician cannot infer that the calcaneal spur is the cause of pain. Very often, inflammation of the soft tissue or bursa beneath the spur gives rise to pain.
- ⊛ Plantar fasciitis, a painful inflammation of the fibrous band of connective tissue (plantar fascia) that runs along the bottom of the foot and connects the heel bone to the ball of the foot. In this condition, the person will have unbearable pain under heel particularly during walking when the weight of the body is carried by the heel.
- ⊛ Tennis elbow is a clinical condition where the patient complains of pain in the lateral epicondyle of the humerus where the extensor group of muscles of forearm is attached. This is also called as lateral epicondylitis. This is more often seen among the tennis players and hence the name, but quite a number of sufferers of this condition have never played tennis. The pain is aggravated when the patient tends to extend the wrist and the fingers, more so against resistance. Diagnosis is done by x ray or MRI.
- ⊛ Carpal tunnel syndrome is a clinical condition which is responsible for the compression of the carpal tunnel. This condition is an example of stenosing tenovaginitis. Here, the cause is not only thickening of the flexor retinaculum, but involvement of the synovial sheaths of the flexor tendons or dislocation of lunate bone which compresses on the contents of this osseo-fibrous canal, median nerve in particular. The main presenting complaint is difficulty in flexing the fingers.

- ⊗ Fibromyalgia is a common nonarticular disorder of unknown cause characterized by generalized aching (sometimes severe); widespread tenderness of muscles, areas around tendon insertions and adjacent soft tissues; muscle stiffness; fatigue and poor sleep. Diagnosis is clinical.
- ⊗ Muscular dystrophies are inherited, progressive muscle disorders resulting from defects in one or more genes needed for normal muscular function. They are distinguished by the selective distribution of weakness and the specific nature of the genetic abnormality involved. Duchenne muscular dystrophy and Becker dystrophy are two common forms of muscular dystrophies observed frequently.

Self-Assessment: Questions for Review

Short essay questions

- ⊗ Describe osteoarthritis
- ⊗ Explain osteoporosis
- ⊗ Describe frozen shoulder syndrome
- ⊗ Describe calcaneal spur
- ⊗ Describe plantar fasciitis
- ⊗ Describe tennis elbow
- ⊗ Describe carpal tunnel syndrome
- ⊗ Explain muscular dystrophy

MCQs

1. Gower's sign is positive in which of the following disease?
 - a. Calcaneal spur;
 - b. Plantar fasciitis;
 - c. Fibromyalgia;
 - d. Duchenne muscular dystrophy

Answer: (d)

2. Which part of the following is afflicted in tennis elbow?
 - a. Medial epicondyle of humerus;
 - b. Lateral epicondyle of the humerus;
 - c. Radius;
 - d. Ulna

Answer: (b)

3. Heberden's nodes are observed in which of the following disease?
 - a. Osteoarthritis;
 - b. Muscular dystrophy;
 - c. Carpal tunnel syndrome;
 - d. Frozen shoulder syndrome

Answer: (a)

4. A 54 years old male individual is diagnosed with the disease osteoarthritis. Which of the following is most likely experiencing?
 - a. Joint swelling;
 - b. Hypermobility of the joint;
 - c. Contracture;
 - d. No swelling

Answer: (a)

5. 60 years old female patient is diagnosed to have suffer-

ing from osteoarthritis. Which of the following can be identified by the clinician on palpation?

- a. Crepitation;
- b. Crepitus;
- c. Heberden's nodes;
- d. A reddened and swollen joint

Answer: (b)

6. Stability in the glenohumeral joint is derived primarily from

- a. Joint contact area;
- b. Ligaments and muscles;
- c. Vacuum in the joint;
- d. None of the above

Answer: (a)

7. Which of the following nerve is compressed in carpal tunnel syndrome?

- a. Radial nerve;
- b. Ulnar nerve;
- c. Median nerve;
- d. Lateral cutaneous nerve

Answer: (c)

8. Dual energy x ray absorptiometry (DEXA) is used in the diagnosis of

- a. Osteoarthritis;
- b. Muscular dystrophy
- c. Frozen shoulder syndrome;
- d. Osteoporosis;

Answer: (d)

9. Cozen's test is performed to diagnose which of the following?

- a. Frozen shoulder syndrome;
- b. Carpal tunnel syndrome;
- c. Calcaneal spur;
- d. Muscular dystrophy

Answer: (b)

10. A clinician views x ray left knee joint of 60 years old male patient and he observes the reduced joint space of the left knee joint with radiolucency on the joint surfaces. He makes the diagnosis of

- a. Osteoarthritis;
- b. Osteoporosis;
- c. Rheumatoid arthritis;
- d. Gout

Answer: (a)

Chapter resources

- ✿ Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- ✿ The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- ✿ Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- ✿ Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- ✿ Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
- ✿ Cecil's medicine edited by Lee Goldman and Dennis Ausiello, 23rd edition, Saunders Elsevier publications, Philadelphia, 2007
- ✿ <https://orthoinfo.aaos.org/en/diseases--conditions/osteoarthritis/>
- ✿ <https://www.sciencedirect.com/science/article/pii/S1063458406003281>
- ✿ <https://www.mdpi.com/1422-0067/24/19/14583>
- ✿ <https://m.indiamart.com/proddetail/frozen-shoulder-treatment-service-19011368012.html>
- ✿ <https://frozenshoulderclinic.com/x-rays/>
- ✿ <https://www.webmd.com/pain-management/heel-spurs-pain-causes-symptoms-treatments>
- ✿ https://en.wikipedia.org/wiki/Calcaneal_spur#:~:text=reduction%20of%20pain.,Prognosis,after%20the%20onset%20of%20symptoms.
- ✿ https://en.m.wikipedia.org/wiki/File:Projectional_radiography_of_calcaneal_spur.jpg
- ✿ <https://www.healthline.com/health/plantar-fasciitis#overview>
- ✿ <https://www.fasciitis.com/plantar-fasciitis-diagnosis/>
- ✿ <https://www.cfaortho.com/media/news/2023/07/tennis-elbow-symptoms-causes-and-treatments>
- ✿ https://www.researchgate.net/figure/Radiograph-of-the-elbow-showing-calcification-of-the-lateral-epicondyle_fig2_291183531
- ✿ <https://mrionline.com/diagnosis/lateral-epicondylitis/>
- ✿ <https://www.facebook.com/physiotherapytool/photos/a.353330381911406/583532298891212/?type=3>
- ✿ https://orthofixar.com/special-test/mills-test/?expand_article=1
- ✿ <https://orthoinfo.aaos.org/en/diseases--conditions/carpal-tunnel-syndrome/>
- ✿ <https://www.sonexhealth.com/patients/carpal-tunnel-syndrome/>
- ✿ <https://en.wikipedia.org/wiki/Fibromyalgia>
- ✿ <https://ordindia.in/duchenne-muscular-dystrophydmd/>
- ✿ <https://kidshealth.org/en/parents/becker-md.html>
- ✿ <https://www.studocu.com/ph/document/lyceum-northwestern-university/physical-therapy/shoulder-complex-mcq/19029246>
